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Research article

Price bubbles and Co-bubbles in the green economy market☆

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ABSTRACT

In light of growing concerns about climate change and environmental issues, investor interest has surged in the new green economy market. However, the existing literature is limited regarding potential price bubbles and co-bubbles within this new domain. This study examines price bubbles and co-bubbles in the new green economy market, covering 31 indexes classified into three groups: the green economy market and its components, geographical regions, and sectors. Using daily data from August 31, 2005, to May 31, 2024, a test procedure is first applied to detect periods of price bubble in the various indexes, then logistic regressions are employed to examine price co-bubble behaviours. The results show evidence of price bubbles in the green economy market, particularly in solar and wind indexes, with peaks during the COVID-19 pandemic and Russia-Ukraine conflict, whereas the water index is the least prone to price bubbles. Regarding geographical region, the USA market exhibits a higher tendency for price bubbles than the Asian or European markets. Several sectors are resistant to price bubbles. The co-bubble analysis reveals a strong reliance of wind index on price bubbles in the solar and water indexes. Price bubbles in Asia significantly influence price bubbles in Europe and the USA. These findings have implications for investment portfolio management and risk management strategies in the new green economy market.

1. Introduction

The energy market plays an essential role in powering the global economy by providing critical inputs for production processes and household consumption. Fossil fuels, particularly crude oil and coal, have historically dominated the energy landscape but their combustion produces substantial carbon dioxide emissions, which contribute to environmental degradation and accelerate climate change (Saeed et al., 2021; Bouri et al., 2022; Wen et al., 2023). This has raised widespread environmental concerns and spurred global initiatives to the transition toward more sustainable energy solutions. In alignment with the 2015 Paris Agreement, governments are increasingly committed to reducing carbon emissions, mitigating climate change, and fostering sustainability in energy production and consumption.

The global response to these environmental challenges has catalysed the rise of a new green economy market and its various indices, tracking the stock performance of various firms closely related to the economic model around sustainable development (Liu et al., 2021; Saeed et al., 2021; Wang et al., 2021). This new green economy market covers environmentally friendly, renewable, and enduring sources such as bioenergy, geothermal, hydropower, solar, and wind, among others. The transition to these energy sources gained even more urgency and visibility during the COVID-19 pandemic, which accelerated global efforts to build a more sustainable economy. Investors have increasingly directed their attention towards firms operating in the renewable energy sector, motivated by growing awareness of environmental issues (Bouri et al., 2022).

However, the clean energy sector is not immune to the broader dynamics of economic cycles and financial market fluctuations. Clean energy assets, though promising in terms of long-term sustainability, are particularly susceptible to volatility and shifts in market sentiment (Zhao, 2020). There are opinions that investor interest in clean

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technology is reminiscent of the "dot-com bubble" (Lehnert, 2022), and will most likely end with an at least equally spectacular crash in the investment market through a practice known as the "painting green" of old traditional instruments (Demidov, 2022). The potential for price bubbles within the green market poses significant risks, as sharp fluctuations in asset prices can lead to market instability, economic disruption, and inflationary pressure (Wang et al., 2024).

The formation of price bubbles, particularly in the clean energy sector, is often tied to period of economic expansion, increased energy demand and perceived supply constraints. These factors can combine to induce an environment where energy prices experience rapid and unstainable growth, leading to volatility across both energy market and broader financial market (Khan et al., 2022). The energy market exhibits significant volatility, particularly in assets associated with clean energy, which frequently experience substantial variation (Rizvi et al., 2022). Various factors, including changes in demand and supply dynamics, geopolitical conflicts, and natural disasters, can cause substantial price swings in the energy market. For example, the COVID-19 pandemic, rising geopolitical tensions in Europe and the Middle East, and the shale gas boom in the United States have all contributed to the oil market's extreme volatility in recent years (Yu et al., 2022).

Several investigations have been conducted to identify the presence of price bubbles in energy commodity markets, notably crude oil (Yang, 2023; Wang et al., 2023; El Montasser et al., 2023) and natural gas (Akcora and Kocaaslan, 2023; Su et al., 2023; Lawal et al., 2022). Wang et al. (2020) study bubbles in Chinese renewable energy sectors, including wind, solar, and hydro, using the generalized supremum augmented Dickey-Fuller (GSADF) test. Their results demonstrate that stock prices in the renewable energy sectors may deviate from their intrinsic values, suggesting the presence of bubbles. However, the literature lacks a thorough analysis of both price bubbles and co-bubbles covering solar, water, and wind indexes, as well as sectoral indexes across geographical regions such as Europe, the USA, and Asia.

This study aims to address this research gap by examining price bubbles and co-bubbles in a large daily datasets of 31 indexes of the new green economy market, covering a wide set of green assets along with their main components, such as solar, water, and wind, and a long period including calm and turbulent periods from August 31, 2005, to May 31, 2024. The GSADF test is used to detect the beginning and ending periods of price bubbles, and logistic regressions are employed to cover co-bubble behaviours. By addressing this research gap, we contribute to a better understanding of the risks associated with the green economy market, offering insights that can be crucial for policymakers, investors, and other stakeholders seeking to navigate the transition to green energy.

Firstly, the bubble testing analysis reveals multiple bubble periods in the green economy market, solar, wind, and water indexes. The green economy index shows frequent bubble periods after 2020, peaking during the COVID-19 outbreak. The solar and wind indexes exhibit the most prolonged bubble periods in 2020–2021, correlating with the pandemic. Wind also reflects the effects of the 2007–2009 global financial crisis (GFC). The global water index is less affected, showing seven bubble periods linked to crises, including the 2008 food price crisis. Geographically, indexes from Asia, Europe, the USA and the rest of the world exhibit distinct patterns in the green economy market and clean energy sector. The analyses show price bubbles in the green economy market and clean energy market indexes across geographical regions. Outside the USA, both indexes exhibit bubbles linked to the 2008 crisis, with additional episodes in 2020. The USA green economy index experiences prolonged periods of price bubble from 2020 to 2022. However, short-lived bubbles occurs in the Asian region. The clean energy index shows similar patterns, with notable bubbles in 2020. The clean energy USA index stands out for its absence of bubbles, warranting further investigation. Sectors including advanced materials, clean energy and developer/operator show price bubbles around the GFC, COVID-19 outbreak, and Paris Agreement.

Secondly, using the date-stamping method, the analyses identify price bubble periods in the green economy market, solar, wind and global water index, with initial occurrences dating back to 2006 and 2007. After 2020, the green economy market index shows prolonged bubble periods coinciding with COVID-19 and geographical events. However, the global water index is not subject to bubbles after 2020 due to decreasing investment opportunities in water energy. Despite some differences, there are overlapping periods of price bubble among the indexes, highlighting the need for further analysis covering co-bubbles in the green economy market, with potential simultaneous price bubbles occurring in multiple indexes for a significant duration. The geographical indexes reveal similarities in the occurrence of price bubbles, notably during the 2007–2008 GFC and post-2020 period. The green economy market index of the USA shows prolonged bubbles after 2020, with clean energy in Asia particularly affected, and three common periods of price bubbles in 2006, 2008 and post-2020, influenced by economic and global health crisis events. Up to four sectors experience simultaneous price bubbles.

Thirdly, logistic regressions reveal that price bubbles in the wind index are highly dependent on the presence of price bubbles in the global water and solar indexes. Co-bubble patterns across geographical regions show that the presence of bubbles in Asia heightens the probability of bubbles in Europe and the USA. Conversely, price bubbles in the Asian clean energy sector affect the occurence of bubbles in its European counterpart. Across sectors, some indexes exhibit minimal dependence on the price bubbles of others, while some indexes, such as green building, natural resources, and transportation, co-bubble more frequently. The renewable energy and fuel cell sectors potentially influence bubbles in various sectors, while pollution mitigation has a negative impact. These findings offer insights into the complex dynamics of price bubbles across sectors and geographical regions within the green economy and clean energy markets.

The subsequent sections of this study are arranged in the following manner: Section 2 offers a thorough overview of the relevant literature; Section 3 provides a detailed explanation of the methodology employed and the data; Section 4 provides the empirical findings; and Section 5 contains the concluding remarks and policy implications.

2. Literature review

The rise of the green economy market reflects a transformative movement towards sustainable practice, emphasizing the integration of environmental responsibility within economic systems (Liu et al., 2021; Chen and He, 2023). This movement is driven by mounting concern over resource scarcity, environmental degradation, and climate change (Bouri et al., 2022; Zhang and Xiang, 2023). Governments, international organizations, businesses, and consumers are increasingly recognizing the urgency of adopting renewable energy sources, eco-friendly technologies, and circular economic models (Ioannidis et al., 2023).

Amidst this movement, concerns have arisen regarding the potential formation of a speculative bubble within the green economy market (Lehnert, 2023). This apprehension stems from the rapid influx of investment into sustainable initiatives, partly fuelled by government incentives, environmental awareness, and heightened consumer demand for eco-conscious products. There is a worry that this rapid investment growth somewhat detaches the market prices of green economy assets from their intrinsic values, making them more driven by speculation and hype than solid economic fundamentals.

Asset price bubbles occur when prices rise rapidly to extremely high levels, and this surge cannot be justified by primary economic factors (Dreger and Zhang, 2013). A significant aspect of financial markets is the occurrence of price bubbles, characterized by rapid and substantial increases in asset prices followed by sharp declines. This price bubble is often associated with speculative behaviour, unsustainable valuations, and market inefficiencies. In recent years, as the world has increasingly emphasized sustainability and environmental consciousness, the emergence of the green economy market has raised questions about the presence of price bubbles and co-bubbles within its components.

In light of classical economic theory, the idea of price bubbles has been thoroughly investigated, especially in the speculative market (Kindleberger, 2005). The Efficient Market Hypothesis (EMH) posits that if the market effectively reflects all available information, price bubbles should not arise, and this should be true in the setting of the green market. However, behavioural finance theories, like those put out by Shiller (2003), contend that investor sentiment and irrational exuberance can cause price deviation from underlying values. The relatively new nature of the green market may contribute to increased price sensitivity and exacerbate bubble tendencies, especially under the effect of external shocks such as the COVID-19 pandemic or geopolitical events and shocks.

The identification of price bubbles is extensively addressed in academic research. Several studies attempt to detect bubble in the prices of crude oil (Yang, 2023; Wang et al., 2023; El Montasser et al., 2023; Wang and Kim, 2022; Pastor and Ewing, 2022; Ajmi et al., 2020), and natural gas (Akcora and Kocaaslan, 2023; Su et al., 2023; Lawal et al., 2022; Umar et al., 2021; Li et al., 2020; Zhang et al., 2018). Henriques and Sadorsky (2008) demonstrate that prices of technology stocks and crude oil independently impact stock prices of alternative energy companies listed on major USA stock exchanges. They highlight the close association between the return behaviour of alternative energy firms and high-tech stocks while emphasizing the limited influence of oil price shocks. Ferrer et al. (2018) examine the dynamics of connectivity in terms of time and frequency between USA clean energy stocks, crude oil prices and important financial factors. They find that there are very few long-term repercussions and that the majority of return and volatility connectivity happens in short term. Creti and Joëts (2017) detect multiple bubbles in European Union emission trading schemes (ETS) from 2005 to 2014. They show distinct periods of price bubbles that coincide with announcements relating to energy and environmental policies. Wei et al. (2022) employ the supremum augmented Dickey–Fuller (SADF) and generalized SADF techniques to identify and analyse the multiple price bubbles present across various major global ETS. They observe that the most significant price bubbles occur within the EU ETS and are strongly associated with energy and environmental occurrences.

Wang et al. (2020) study bubbles in specific Chinese renewable energy sectors such as wind, solar, and hydro using GSADF. Combined with a bubble model, the findings show that stock prices in renewable energy sectors could differ from their actual values, indicating bubble-like behaviour. These tendencies may be attributed to various factors, including trade barriers, economic growth, global financial crises, and environmental pressure for protection and sustainability. Kassouri et al. (2021) explore whether oil, clean energy, and high technology stock prices are closely related, considering speculative bubbles. Their findings indicate that a shock disrupting the oil market, resulting in increased oil prices, significantly enhances investment in clean energy. This upsurge primarily stems from the motivation to substitute fossil fuels with cleaner energy sources. Lehnert (2023) finds multiple bubbles in the green stock market, employing a recursive testing procedure. The findings show speculative bubbles and a significant surge in that market beginning in June 2021, potentially linked to the advent of new green technology. Memon et al. (2023) apply the multifractal detrended fluctuation method to analyse herding behaviour within clean and renewable energy markets. They scrutinize this behaviour before and during the COVID-19 period. Their analysis indicates multifractality within these markets and notably emphasizes an augmented symmetry in herding behaviour, particularly during the crisis.

Many studies test potential price bubbles in financial markets. The asset pricing model of Lucas (1978) provides the foundational framework for examining rational bubbles arising when asset prices diverge from their fundamental values. LeRoy and Porter (1981) use the variance bounds test to pinpoint bubbles in Standard and Poor's stock composite index. Blanchard and Watson (1982) examine rational bubbles in financial markets using "runs tests" and "tail tests". West (1987) finds speculative bubbles in Shiller's dataset by employing ARIMA and the Euler equations (two-step test). Diba and Grossman (1988) use cointegration tests to examine rational bubbles. Bohl (2003) applies the Enders–Siklos momentum threshold autoregressive approach to analyse bubbles in the USA stock market. The above methods used to identify price bubbles often produce unconvincing and questionable results, which raises concerns regarding their validity and reliability. Many studies rely on the unit root test, which may not be suitable for bubble detection due to its low efficacy and the plausible existence of bubbles (Evans, 1991). Additionally, the Markov switching ADF might not correctly identify when bubbles start or reach their highest points (Shi, 2010). However, Phillips et al. (2015) highlight the enhanced effectiveness of the GSADF tests compared to other econometric methods. The GSADF tests stand out for their proficiency at capturing dynamic movements across an entire dataset, ensuring a rich pool of observations for an adequate estimation process. An inherent advantage lies in their ability to handle multiple exuberance instances and subsequent collapses. With their broader coverage of data subsamples and enhanced window flexibility, these tests are specifically engineered to outperform in instances of bubbles behaviour occurring multiple times within the data. Several studies apply GSADF to identify bubble periods in asset prices (see, among others, Caspi et al., 2018; Bouri et al., 2019; Pastor and Ewing, 2022; Shahzad et al., 2022).

According to the literature reviewed above, no previous studies examine price bubbles in solar, water, and wind indexes, along with sectoral indexes across various geographical regions such as Europe, the USA, and Asia, in addition to price co-bubbles. Such a research gap significantly hampers our understanding of the potential economic risks and opportunities inherent to this vital green economy market. Therefore, this study addresses this research gap by examining price bubbles in the green economy market using the GSADF method, and price co-bubbles using logistic regressions. The primary benefit of using Philip's methodology (Phillips et al., 2015) rests in its adequate framework, which can accommodate diverse structural breaks and fluctuations in data dynamics over time. This adaptability renders these tests versatile and well-suited for various market conditions. This technique is ideally suited for analysing extensive time series, such as our case spanning August 31, 2005, to May 31, 2024, and thus the dynamic nature of price bubbles in the green economy market.

3. Data and methodology

3.1. Data

This study considers daily data on 31 indexes representing the green economy market, published as part of the NASDAQ OMX Green Economy Family of indexes.[1] Appendix A provides a description of the 31 indexes under study.[2] The sample period is August 31, 2005 to May 31, 2024, with a daily frequency of each index, covering calm and crisis periods. The beginning of the research period is determined by data availability. All used indexes are denominated in USD. The 31 indexes are divided into three main groups. The first involves the green economy market, divided into wind, solar, and water indexes; bearing in mind that the Main Green Economy Market Index represents the entire markets in this group. The next group considers the geographical criterion, in which two categories of indexes are distinguished from the green economy market and clean energy families. The green economy market

[1] We use price indexes for the main analysis, whereas the total return indexes are used for robustness checks (see Section 5).

[2] The list of all companies constituting each index is available in NASDAQ methodology files, including information about company names, weights, exchanges, and currency.

focuses on environmentally sustainable practices and technologies aiming to reduce carbon footprints and promote resource efficiency. The clean energy market specifically targets the production and distribution of energy from renewable sources such as solar, wind, and hydro, striving to replace fossil fuels and reduce greenhouse gas emissions. For both of these categories, indexes covering Asia, Europe, USA, and outside USA are used. The third group considers the market examined based on 19 sectors. Appendix Table B (Table B.1) presents the summary statistics of the 31 indexes under study.

3.2. Methodology

Following Phillips et al. (2015), we use the generalized supremum augmented Dickey-Fuller (GSADF) test to identify the origination and termination dates of periods of price bubbles:

$$GSADF(r_0) = \sup_{\substack{r_2 \in [r_0,1] \\ r_1 \in [0, r_2 - r_0]}} ADF_{r_1}^{r_2} \tag{1}$$

where:

r_0 – minimum length of the test window,

r_1 – start point of the test window,

r_2 – end point of the test window,

ADF – the value of the statistic for an augmented Dickey-Fuller test. This is standard ADF unit root testing against stationarity.

The standard ADF regression is given by:

$$\Delta y_t = \alpha_{r1,r2} + \beta_{r1,r2} y_{t-1} + \sum_{i=1}^{k} \Psi_{r1,r2}^{i} \Delta y_{t-i} + \varepsilon_t \tag{2}$$

where:

y – index value (each index from the NASDAQ OMX Green Economy Family),

k – lag order,

α, β, Ψ, – estimated parameters values,

ε – error term.

Based on Equation (2), the SADF formula is:

$$SADF(r_0) = \sup_{r_2 \in [r_0,1]} ADF_0^{r_2} \tag{3}$$

which is linked to the GSADF equation as follows:

$$GSADF(r_0) = \sup_{r_2 \in [r_0,1]} SADF(r_0) \tag{4}$$

The GSADF test depending on the test window parameters is presented schematically in Fig. 1.

The GSADF test is superior to earlier versions of the test, such as SADF and ADF, because, as presented in Fig. 1, it relies on various test window lengths (ADF uses only the length equal to the sample interval)

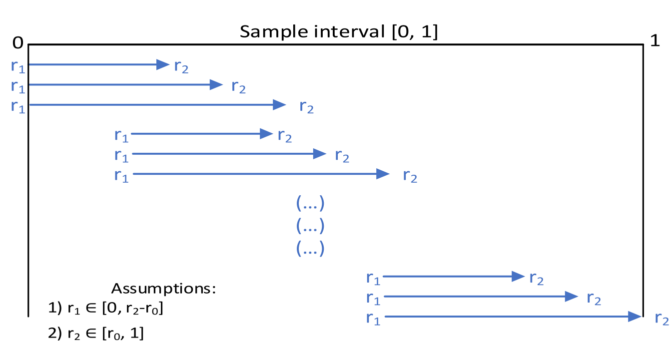

Fig. 1. Graphical presentation of GSADF test parameters.
Source: Own study, based on the work of Phillips et al. (2015).

and various values for the start test window, i.e. r_1 (SADF uses only r_1, always equal to 0, with a set sample interval). This testing procedure makes the GSADF test advisable for testing long time series and those in which multiple periods of price bubbles may occur. Such properties of the GSADF test are pertinent to our analysis covering daily data for a long period of almost twenty years.[3]

The values resulting from the described testing procedure are compared to the critical values, which are obtained using Monte Carlo simulations with 1000 repetitions.[4]

To date-stamp the beginning and ending of the price bubble periods, the backward SADF (BSADF) statistic is calculated as:

$$BSADF_{r2}(r_0) = \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2} \tag{5}$$

Then, we examine co-occurrence of price bubbles or price co-bubbles across indexes using logistic regressions. To this end, the value of 1 is given for the day on which price bubble is detected, and zero otherwise. Notably, we analyse price bubble only when the length of the detected period of bubble is equal to at least three days (Etienne et al., 2014). Finally, our general model equation for catching co-bubble between detected price bubbles is:

$$\log\left(\frac{P(Y=1|X)}{1 - P(Y=1|X)}\right) = \beta_0 + \beta_i {}^* X_{i,t} + \varepsilon_t \tag{6}$$

where,

Y – a dependent variable indicating if a price bubble is detected,

$X_{i,t}$ – a set of dummy variables (the number of variables depends on the groups, categories, or sectors under study),

β_0 – constant term,

β_i – coefficient value,

ε_t – error term.

The methodology used to examine co-bubble is similar to previous studies (Bouri et al., 2019; Shahzad et al., 2022). The estimated results report the McFadden R^2 statistic (McFadden, 1973) to show how well the obtained model fits the datasets.

4. Research results

4.1. Detection of price bubbles periods

The GSADF test results are visualized in Fig. 2, where the calculated test statistic values are shown, with the dashed line representing the critical test value. The shaded areas represent bubble periods, indicating periods where the calculated test statistic exceeds the critical value. The presence of significant price bubble periods suggests irrational investing (Baker and Ricciardi, 2014; Hirshleifer et al., 2006).

Several bubble periods can be identified from Fig. 2 for each data series examined. The first, in the main green economy market index, occurs from May 8, 2006 to May 12, 2006. Two weeks before this period of bubble, the main index rose nearly 4%, reached a peak on May 9, 2006, then dropped by more than 3.5% in the next four days. After this, the main index shows four short-lived bubble periods, not exceeding 7 days, from October to November 2008. Overall, the most prolonged and frequent occurrences are observed for the main index after 2020. Then, seven price bubble periods are observed, lasting from 3 to 71 days. During these periods the index spikes each time, on average, by approximately 8%, with the highest rise exceeding 15% two times from December 3, 2020 to January 8, 2021, then from October 7 to November 4, 2021. These bubble periods coincide with the COVID-19 outbreak period and associated development of the pandemic. Previous studies

[3] Price bubble and price explosivity are often used interchangeably (Xu and Salem, 2021).

[4] All model estimations performed in this study are done using the R program, specifically the "exuber" package (Vasilopoulos et al., 2022).

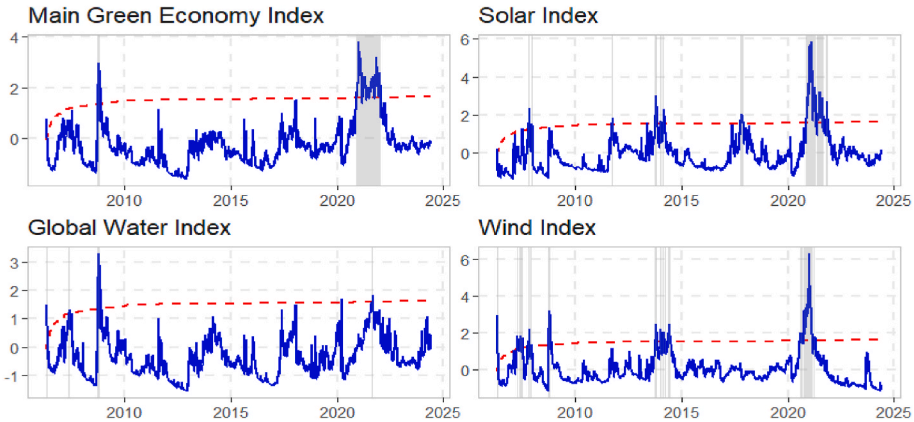

Fig. 2. Price bubble in the green economy market and its components. Notes: The blue line shows the individual price index, and the shaded areas represent bubble episodes. The dashed line indicates the critical value. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

indicate the presence of bubble phases for the main green economy market index around the COVID-19 outbreak (Basse et al., 2023).

Quite similar results are obtained for the solar and wind indexes, where the longest periods of price bubble is detected in 2020. For the solar index the longest price bubble period starts on November 19, 2020 and lasts for 116 days until May 10, 2021. In 2021 four more bubble periods were detected with, on average, more than a 12% increase in the index value. Evidence of COVID-19 impacting the green economy market sectors more than the 2007–2009 GFC is also observed by Demiralay et al. (2023). For the wind index, the longest price bubble period (105 days) starts in September 29, 2020, reaching its peak in January 7, 2021 and ending on March 2, 2021. However, for the wind index nine bubble periods are noted in 2007 and 2008, which indicates the strong impact of the GFC on that index. The less affected index in the group of main green economy indexes is the global water index, for which the main price bubble periods are observed around the GFC, with a short period of price bubble noted around the COVID-19 pandemic. Such results could be explained by the food price crisis which began in 2008, affecting water resources (Winpenny et al., 2009).

Moving to the indexes constructed based on geographical criteria,

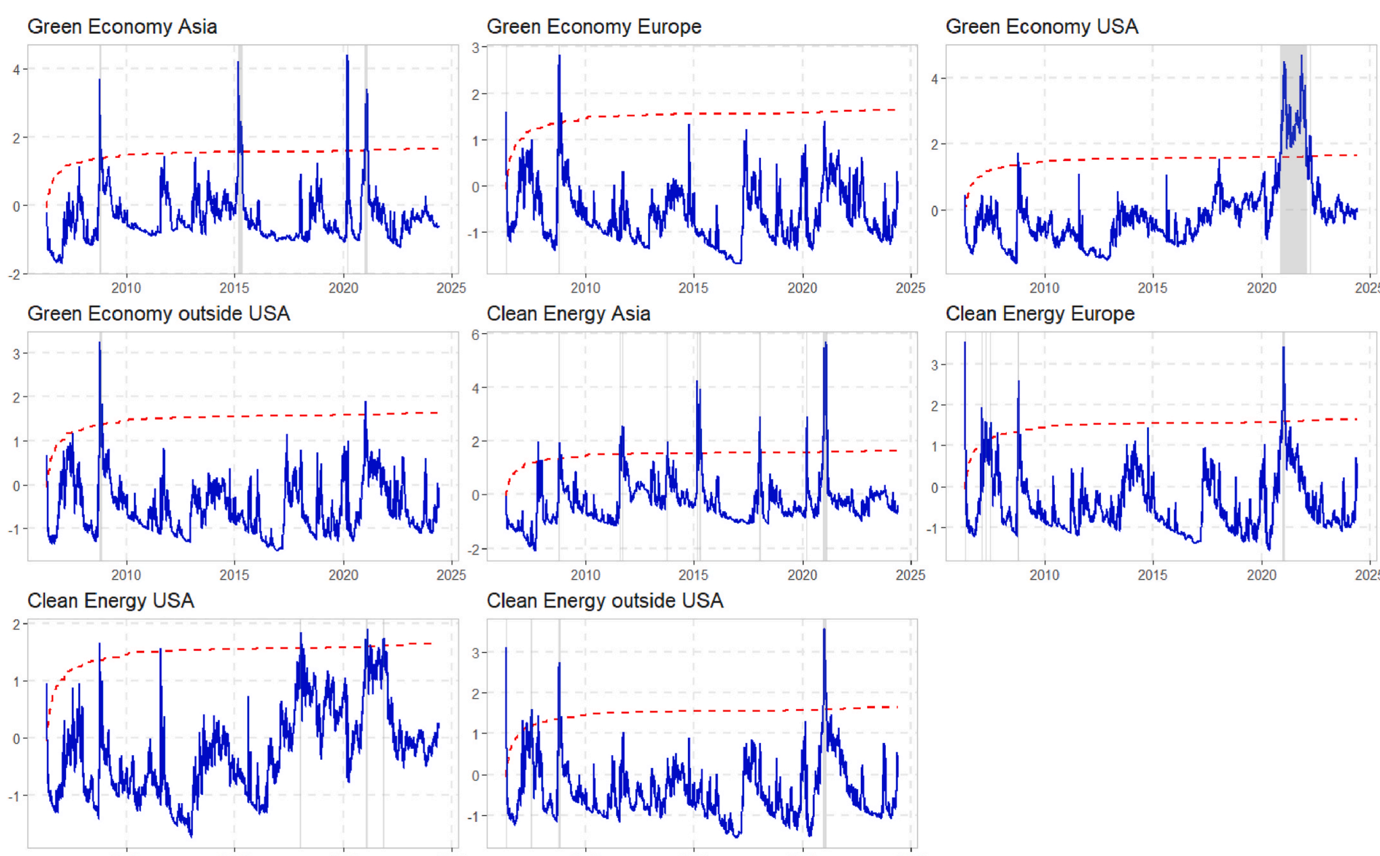

Fig. 3. Price bubble in green economy market and clean energy markets based on geographical areas. Notes: The blue line shows the individual price index, and the shaded areas represent bubble episodes. The dashed line indicates the critical value. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

covering Asia, Europe, the USA, and the rest of the world excluding the USA, they are divided into two groups, green economy market and clean energy market. The GSADF test results for these geographical regions are presented in Fig. 3.

We note three price bubble periods for the green economy market index outside USA, between 2006 and 2008, around the GFC. The highest price increase is almost 10% and the biggest decline exceeds 15%. Almost identical results emerge for the clean energy index outside USA. Price bubbles also occur between 2006 and 2008, with one additional period in 2020, coinciding with the COVID-19 pandemic. The recorded price changes range from 27% to −28%. For the green economy market USA index, a price bubble lasts 299 days, from November 18, 2020 until January 27, 2022, during which time the index value first increases more than 62%, then, after its peak on November 4 in 2021, declines by more than 20%. Such large and rapid changes should be monitored because the USA is the world's biggest contributor to the green economy (1.3 trillion USD annual sales and 9.5 employees) according to Georgeson and Maslin (2019). During the same period, three short-lived periods are observed in the Asia region, however no such periods occur in Europe in the green economy market indexes. The longest periods of price bubble in the clean energy indexes are noted around 2020, agreeing with Gunay et al. (2022). The clean energy USA index shows no price bubble at a significance level of at least $\alpha = 0.05$ (this is true for both the price and total return versions of the indexes). Accordingly, the clean energy USA index is excluded from further analysis.

In the last category, the indexes are analysed according to their specific sectors, and the results are presented in Fig. 4. Three price bubble periods are observed for the advanced materials index. The first lasts for 10 days, from May 8, 2006 to May 22, 2006, the second for 16 days from July 3, 2007 to July 26, 2007, and the last for 4 days in 2017. Eight price bubble periods are noted for the clean energy index, five in 2006–2008 around the GFC, and three around 2021, which can be linked to the COVID-19 pandemic. Twice in 2008, during these periods, the index value increases by more than 15% and decreases by more than 12%. The same concentration of price bubbles is found for the developer/operator index, with ten periods observed in 2006–2008 and three in 2020. A different situation is seen for the fuel cell index, which experiences eight periods in 2020 and only one in the period 2006–2008. Overall, the same pattern is present for the rest of indexes analysed, such as price bubbles mainly are present in the period 2006–2008 and from 2020. The association between the COVID-19 pandemic and the price behaviour of the selected indexes from the green economy market is also mentioned by Zeng et al. (2023). We link the high number of price bubble periods occurring after 2020, not only to COVID-19, but also to the fact that climate change forces changes in global investment in clean energy. The growth in the green economy market investment segment is driven by key events such as the adoption of the United Nations 17 Sustainable Development Goals in September 2015 (Gunay et al., 2022) or Paris Agreement regulations which started at the end of 2016. Such climate meetings and agreements are the basis for new investment opportunities which ultimately constitute the source of the price bubble periods from 2020 onwards (Zeng et al., 2023; Ferreira and Morais, 2022; Hassan, 2022).

Furthermore, we do not find evidence of the existence of multiple price bubbles for the six sectors green IT index (GREENIT), bio/clean fuels index (GRNBIO), energy efficiency index (GRNENEF), smart grid index (GRNGRD), lighting index (GRNLIGHT) and energy management (GRNMAN). Therefore, no further analysis is conducted of these indexes. We link the resistance for those sectors to price bubbles with their good hedging abilities for traditional carbon and energy stock companies (Sun et al., 2024). Green economy sectors are mainly net transmitters of shocks to other markets in times of stress and this is possibly why such price bubbles do not affect the above sectors. The ability of a green economy market to serve as a hedge for other investments is also noted in Chen et al. (2023). However, those suggested reasons require an empirical examination to ensure their validity, possibly using multi-layer connectedness networks.

4.2. Date-stamping analysis

Fig. 5 shows a timeline of the occurence of price bubbles in the green economy market index and its main components (solar, water, and wind). This date-stamping analysis is the basis for conducting co-bubbles analysis.

Fig. 5 shows the initial occurrence of price bubble in the green economy market index from May 8 to May 12, 2006, a pattern mirrored in the global water and wind indexes, representing an overlapping period. However, the solar index sees its first episode of price bubble from October 29 to November 12, 2007, then again between December 24, 2007 and January 4, 2008. The most prolonged periods of price bubble for the green economy market index are observed after 2020, coinciding with the COVID-19 pandemic and the Russian aggression against Ukraine. Notably, there is a high visual co-occurrence of price bubble between the main green economy market index and the solar index. Conversely, for the wind and global water indexes and the main green economy market index, the visual co-occurrence of periods of price bubble is less evident. The global water index has the fewest instances of price bubble, and this index remains unaffected by price bubble after 2020. We link the occurrence of price bubble in the global water index after 2007 with the food crisis (Winpenny et al., 2009). Conversely, the inability of the global water index to experience price bubble periods after 2020 could be due to the decreasing role of energy from water in the past decades. This is associated with fewer investing opportunities in that sector (Moore and Henbest, 2020).

In Fig. 6, we present the results of the data-stamping procedure as a preliminary assessment of the co-occurrence of diagnosed price bubble. We focus only on indexes for which, based on the GSADF test, the occurrence of price bubble is confirmed.

As shown in Fig. 6, the geographical indexes also show some similarities in price bubbles. Firstly, during the crisis period of 2008, each index experiences short periods of price bubble. Secondly, excluding the green economy market outside the USA and green economy market Europe, most extended periods of price bubbles are observed after 2020, which is extremely visible for green economy market USA. Finally, the clean energy Asia index has the largest number of price bubble periods. An overlap exists between the price bubble periods in the indexes analysed. In the green economy market indexes, price bubbles occur simultaneously almost 1.3% of the time for clean energy indexes.

Fig. 7 shows periods of price bubble on a timetable for the sectors analysed. Notably, three common periods, when price bubble occurs in the sectors, are indicated. The first is in 2006, when 11 sectors experience price bubbles. For nine of these sectors, advanced materials, clean energy, fuel cell, green building, healthy living, natural resources, recycling, renewable energy and transportation, the first period of price bubble occurs in May; for green building in November and for the developer/operator sector in December. The second period of price bubble for almost all sectors is from October to November 2008. The last such period is after 2020, when we see the longest such periods. Overall, the visual co-occurrence of price bubble periods is a reason for further analysis involving logistic regressions to study co-bubbles.

4.3. Price bubbles characteristics

Table 1 provides a summary of the episodes of price bubbles, their average duration, the maximum duration measured in days, and percentage of time with price bubbles.

The results for the main green economy market index and the solar and wind indexes, in Panel A, indicate that the total time during which there are price bubbles is 327 days, more than 7.5% of the time studied. The total number of price bubble periods is from 12 (main green economy market index) to 22 (wind index). For the global water index, we

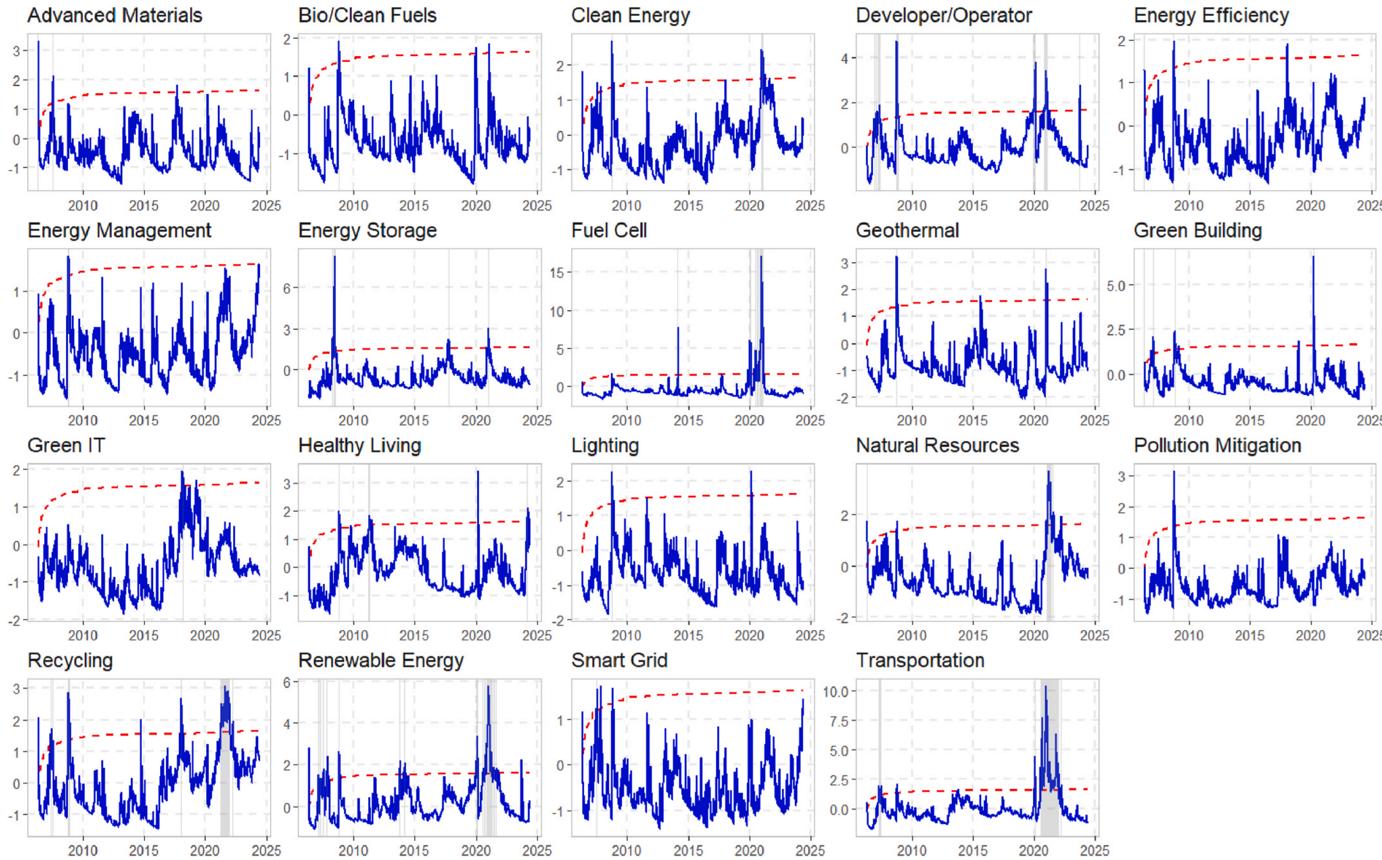

Fig. 4. Price bubble in the green economy market and clean energy markets based on various sectors. Notes: The blue line shows the individual price index, and the shaded areas represent bubble episodes. The dashed line indicates the critical value. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

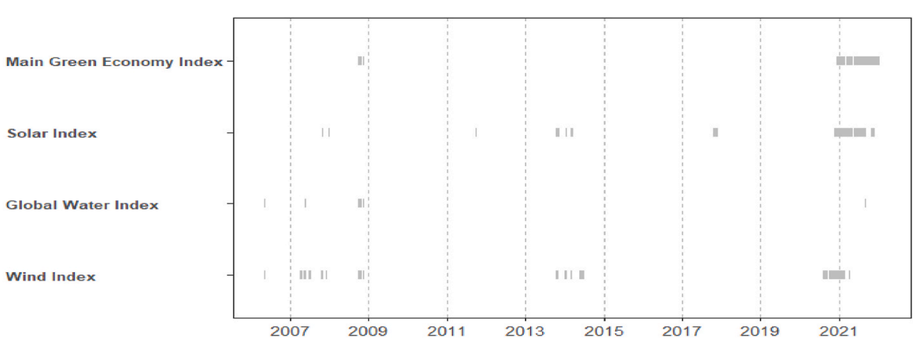

Fig. 5. Periods of price bubble in the green economy market and its components. Notes: A period is considered bubble if it persists for a minimum of three consecutive days. The shaded areas indicate the occurrence of bubbles.

observe fewer (7) and shorter (average 5 days) periods than the other indexes. Significantly, this reflects the lack of impact of the COVID-19 pandemic on the price behaviour of the global water index. According to Moore and Henbest (2020) wind and solar sources of energy lead the global electricity generation mix, with a decreasing role of energy from water. This is additional factor can induce price bubble periods in solar and wind indexes. Panel A ranks the indexes in terms of the number of price bubble periods: wind, solar, water.

Panel B indicates that the index with the longest price bubbles is the green economy market USA, with price bubbles lasting 323 days. This indicates price bubble periods equal to 7.4% of the period analysed. Thus, the green economy market index in the USA is more susceptible to price bubbles than the indexes covering Asia or Europe. Importantly, for the remaining indexes, the diagnosed periods of price bubble last an average of twelve days, which is a short duration compared to the findings from previous studies (Wei et al., 2022; Lehnert, 2023). Panel B also shows that there are more periods of price bubble for the clean energy focused indexes than the green economy market, comparing the same regions with statistically significant results.

Panel C indicates that a group of sectors containing advanced materials, clean energy, green building, geothermal, healthy living, pollution mitigation and energy storage, exhibit short (on average not exceeding 15 days) and small numbers (equal to or less than 9) of detected price bubble periods. Another group of sectors, comprising developer/operator, fuel cell, natural resources, recycling, renewable energy, and transportation, have price bubble periods lasting more than 130 days, which is more than 3% of the time studied. This division of sectors could be valuable for investors in term of portfolio diversification

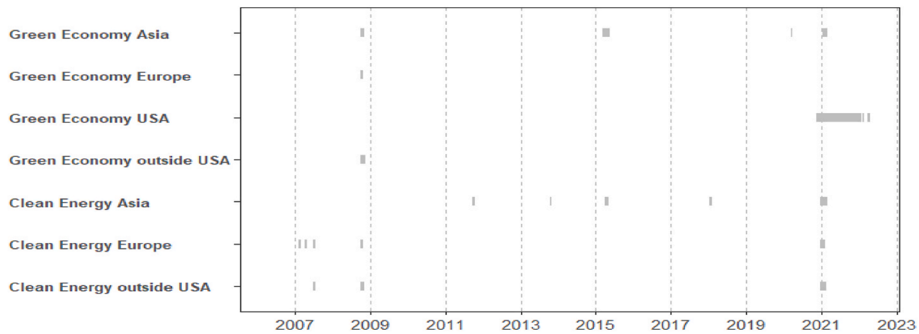

Fig. 6. Periods of price bubble for geographical areas. Notes: A period is considered bubble if it persists for a minimum of three consecutive days. The shaded areas indicate the occurrence of bubbles.

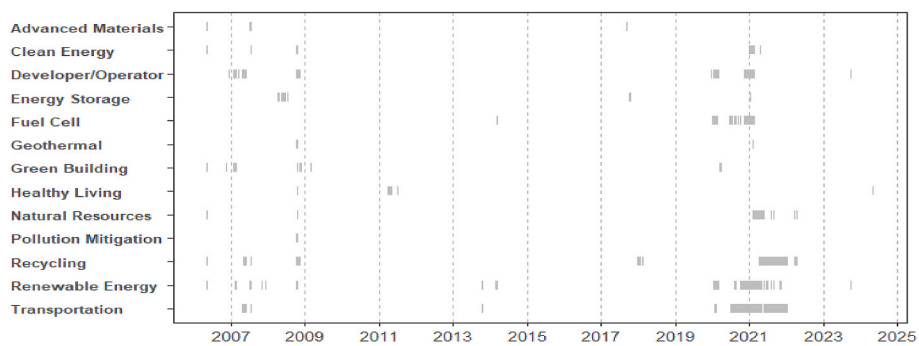

Fig. 7. Periods of price bubble for sectoral analysis. Notes: A period is considered bubble if it persists for a minimum of three consecutive days. The shaded areas indicate the occurrence of bubbles.

Table 1
Characteristics of price bubble periods.

Index	Length sum [days]	Maximum length [days]	Number of price bubbles	Standard deviation length [days]	Average length [days]	% time of price bubbles
Panel A: Price bubble for the main green economy index and its components						
Main Green Economy Index	296	71	12	26	25	6.77%
Solar Index	309	116	15	30	21	7.07%
Global Water Index	38	7	7	2	5	0.87%
Wind Index	327	105	22	21	15	7.48%
Panel B: Price bubble based on geographical area						
Green Economy Asia	114	39	7	11	16	2.61%
Green Economy Europe	27	12	4	4	7	0.62%
Green Economy USA	323	299	3	166	108	7.39%
Green Economy Outside USA	33	21	3	9	11	0.75%
Clean Energy Asia	143	53	11	15	13	3.27%
Clean Energy Europe	95	31	8	9	12	2.17%
Clean Energy Outside USA	81	38	7	12	12	1.85%
Panel C: Price bubble based on sectors						
Advanced Materials	30	16	3	6	10	0.69%
Clean Energy	69	34	8	10	9	1.58%
Developer/Operator	237	70	14	19	17	5.42%
Energy Storage	103	36	7	11	15	2.36%
Fuel Cell	190	79	10	23	19	4.34%
Geothermal	25	9	5	3	5	0.57%
Green Building	82	19	8	7	10	1.88%
Healthy Living	52	13	9	3	6	1.19%
Natural Resources	133	76	10	22	13	3.04%
Pollution Mitigation	20	8	4	2	5	0.46%
Recycling	331	200	9	62	37	7.57%
Renewable Energy	372	129	27	24	14	8.51%
Transportation	462	216	14	67	33	10.56%

Notes: This table provides information on price bubble lasting for at least 3 days. Panel A gives a summary of the green economy market index and its components, Panel B gives statistics on bubbles based on geographical area, and Panel C gives information based on sector.

Table 2
Logistic regression results – components of the main green economy index.

	Solar Index	Global Water Index	Wind Index
Solar Index		−1.439 (0.548) ***	2.640 (0.140) ***
Global Water Index	−1.439 (0.548) ***		4.334 (0.410) ***
Wind Index	2.640 (0.140) ***	4.334 (0.410) ***	
Const (β_0)	−3.098 (0.074) ***	−6.276 (0.354) ***	−3.141 (0.076) ***
McFadden R2	0.131	0.303	0.177

Notes: This table presents the results of co-bubble across the components of the main green economy index, namely the solar, water, and wind indexes. ***, **, and * denote statistical significance at the 0.01, 0.05 and 0.10 levels, respectively.

in times of stress. Our classification is broader, covering more indexes than the existing literature, but is consistent with previous findings (e.g. Kuang, 2021).

4.4. Evidence of co-bubbles

Firstly, we estimate the results of price co-bubble across the components of the main green economy index, namely the solar, water, and wind indexes,[5] presented in Table 2. The results indicate that the bubble in the solar index is highly dependent on the presence of bubble in the global water and wind indices. Price bubble in the wind index significantly heightens the likelihood of price bubble in the solar index. Moreover, the existence of price bubble in the global water index decreases the likelihood of having price bubbles in the solar index. However, bubble in the water index has a positive effect on the occurrence of bubble in the wind index.

Secondly, we investigate co-bubble patterns across geographical regions, with detailed results presented in Table 3. NASDAQ provides distinct sets of geographical indexes related to the green economy market and clean energy market. We segregate these indexes into two separate panels (A and B). The results for the green economy in Panel A show that the occurrence of price bubbles in Europe and the USA elevates the likelihood of price bubbles in the green economy market of Asia. However, price bubble in Europe and the USA is contingent solely on price bubbles of the green economy market in Asia. This suggests that the presence of bubbles in Asia heightens the probability of bubbles in Europe and the USA. We connect those interesting geographical dependencies with Asia's involvement in green financing. The scale of it, comparable to USA and Europe, can impact the green economy markets in those regions (Lichtenberger et al., 2022). Additionally, in recent years, the trend of intellectual property formation has shifted towards Asia, especially in innovative practices in Asia connected with the green economy sector (Tang et al., 2023). The clean energy results in Panel B show that the presence of bubbles in Europe increases the likelihood of bubbles in the clean energy sector of Asia, and vice versa.[6] Notably, price bubbles in Europe are solely dependent on price bubbles of clean energy in Asia.

Thirdly, we examine co-bubbles across various sectors and present the results in Table 4. Geothermal and pollution mitigation sectors have minimal dependence on price bubbles in other sectors. In contrast, prices of the renewable energy and fuel cell sectors tend to co-explode more

[5] Given that the solar, water and wind indices are components of the main green economy index, the main green economy index is excluded from the co-bubbles analysis.

[6] The coefficient for the green economy Europe index is insignificant in the first-step model due to its high p-value, which accounts for its removal from the subsequent backward stepwise regression. For the clean energy USA index, the GSADF test does not detect any periods of price bubbles.

Table 3
Logistic regression results for geographical indexes cross-section.

	Green Economy Asia	Green Economy Europe	Green Economy USA	Clean Energy Asia	Clean Energy Europe
Panel A: Green Economy					
Green Economy Asia		4.225 (0.405) ***	1.711 (0.219) ***		
Green Economy USA	1.711 (0.219) ***				
Panel B: Clean Energy					
Clean Energy Asia					3.252 (0.233) ***
Clean Energy Europe				3.252 (0.233) ***	
Const (β_0)	−3.954 (0.111) ***	−6.038 (0.302) ***	−2.696 (0.06) ***	−3.746 (0.098) ***	−4.341 (0.131) ***
McFadden R2	0.043	0.257	0.02	0.108	0.149

Notes: This table presents the results of co-bubble in the main green economy index and clean energy market across geographical areas. ***, **, and * denote statistical significance at the 0.01, 0.05 and 0.10 levels, respectively.

frequently. Specifically, bubbles in the healthy living sector increase the chance of bubbles in the geothermal sector, while price bubbles in pollution mitigation are affected by green building, healthy living, renewable energy, and transportation price bubbles. The results also show that bubbles in the clean energy and renewable energy sectors raise the likelihood of bubbles in the advanced materials sector. However, price bubbles in the healthy living and natural resources sectors decrease the occurrence of price bubbles in the developer/operator sector. Interestingly, bubbles in the energy storage, fuel cell, green building, recycling, and renewable energy sectors boost the probability of bubbles in the developer/operator sector. For clean energy, the occurrence of price bubbles increases when there are price bubbles in the developer/operator, energy storage, fuel cell, natural resources, recycling, and renewable energy sectors. The likelihood of price bubbles in the energy storage sector increases when there are price bubbles in the clean energy and fuel cell sectors, but decreases in the presence of bubbles in the renewable energy and transportation sectors. The likelihood of bubbles in the healthy living sector rises with the presence of bubbles in the clean energy and green building sectors, but decreases with price bubbles in the developer/operator sector. The occurrence of bubbles in the renewable energy sector is positively influenced by the presence of bubbles in the advanced materials, clean energy, developer/operator, fuel cell, green building, natural resources, and transportation sectors, but it is adversely affected by bubbles in the recycling sector. The occurrence of price bubbles in the fuel cell sector increases with the presence of price bubbles in the developer/operator, healthy living, renewable energy, and transportation sectors. Conversely, price bubbles in the recycling and natural resources sectors have a negative impact.

The likelihood of price bubbles in the natural resources sector increases when there are price bubbles in the clean energy, green building, healthy living, recycling, renewable energy, and transportation sectors. Conversely, the formation of price bubbles in the natural resources sector decreases with the presence of price bubbles in the advanced materials, developer/operator, and pollution mitigation sectors. The potential for price bubbles in the recycling sector is positively influenced by bubble behaviours in the prices of the advanced materials, clean energy, developer/operator, geothermal, green building, natural resources, and transportation sectors. However, it is adversely affected by price bubbles in the fuel cell and renewable energy sectors.

Green building has heightened vulnerability to price bubbles, as

Table 4
Logistic regression results for sectoral indexes.

	Advanced Materials	Clean Energy	Developer/ Operator	Energy Storage	Fuel Cell	Geothermal	Green Building	Healthy Living	Natural Resources	Pollution Mitigation	Recycling	Renewable Energy	Transportation
Advanced Materials							1.653 (0.839) **		−5.446 (1.119) ***		2.725 (0.673) ***	4.133 (0.557) ***	−3.616 (1.258) ***
Clean Energy	2.425 (0.484) ***			4.814 (1.081) ***			2.368 (1.079) **	2.769 (0.547) ***	3.864 (0.822) ***		2.197 (1.253) *	4.598 (1.11) ***	2.429 (1.318) *
Developer/ Operator		3.947 (0.72) ***			1.705 (0.302) ***		3.889 (0.348) ***	−1.535 (0.635) **	−2.425 (0.768) ***		1.675 (0.349) ***	2.544 (0.271) ***	
Energy Storage		6.387 (2.753) **	1.276 (0.432) ***			3.655 (0.891) ***							
Fuel Cell	−2.064 (0.661) ***	2.8 (1.074) ***	2.786 (0.261) ***	1.912 (0.513) ***		2.077 (0.852) **	−2.723 (0.914) ***				−10.364 (1.322) ***	2.512 (0.306) ***	3.224 (0.299) ***
Geothermal							−3.872 (1.166) ***				4.211 (0.963) ***		
Green Building			3.634 (0.296) ***					3.109 (0.413) ***	1.49 (0.824) *	1.977 (0.603) ***	2.15 (0.379) ***	1.68 (0.388) ***	−2.438 (0.687) ***
Healthy Living			−2.303 (0.833) ***		2.101 (0.671) ***	2.313 (0.872) ***	3.437 (0.413) ***		1.857 (0.882) **	3.031 (0.87) ***			1.746 (0.611) ***
Natural Resources		1.731 (0.466) ***	−1.567 (0.335) ***		−1.312 (0.381) ***		1.325 (0.687) *				3.014 (0.365) ***	3.077 (0.329) ***	0.66 (0.312) **
Pollution Mitigation									−3.913 (1.152) ***				−7.365 (1.437) ***
Recycling		5.629 (1.191) ***	1.73 (0.241) ***		−4.615 (0.668) ***	4.378 (0.814) ***	0.819 (0.435) *		2.71 (0.321) ***			−0.63 (0.269) **	4.881 (0.203) ***
Renewable Energy	3.224 (0.493) ***	5.23 (1.148) ***	2.322 (0.228) ***	−2.552 (1.125) **	2.44 (0.312) ***	4.192 (0.777) ***	0.834 (0.433) *		3.085 (0.318) ***	4.302 (0.648) ***	−1.794 (0.294) ***		3.777 (0.243) ***
Transportation				−1.096 (0.623) *	3.713 (0.289) ***	−3.091 (0.614) ***	−3.212 (0.792) ***		0.901 (0.378) **	−2.793 (0.852) ***	5.11 (0.216) ***	3.402 (0.241) ***	
Const (β_0)	−6.294 (0.354) ***	−13.462 (1.786) ***	−4.44 (0.138) ***	−3.892 (0.109) ***	−5.21 (0.196) ***	−8.526 (0.758) ***	−5.005 (0.186) ***	−4.813 (0.165) ***	−5.759 (0.245) ***	−7.053 (0.482) ***	−4.122 (0.122) ***	−4.5 (0.142) ***	−4.453 (0.142) ***
McFadden R2	0.311	0.719	0.421	0.069	0.626	0.540	0.328	0.130	0.543	0.410	0.546	0.617	0.652

Notes: This table presents the results of co-bubble in the green economy index and clean energy market across geographical areas. ***, **, and * denote statistical significance at the 0.01, 0.05 and 0.10 levels, respectively.

evidenced by its significant co-movement with ten other sectors. The occurrence of price bubbles in the green building sector increases in the presence of price bubbles in the advanced materials, clean energy, developer/operator, healthy living, natural resources, recycling, and renewable energy sectors. Conversely, price bubbles in the fuel cell, geothermal, and transportation sectors have a negative effect. The existence of price bubbles in the clean energy, fuel cell, healthy living, natural resources, recycling, and renewable energy sectors augments the likelihood of price bubbles in the transportation sector. Conversely, the presence of price bubbles in the advanced materials, green building, and pollution mitigation sectors reduces the likelihood of price bubbles in the transportation sector. Taken together, the results show that sectors exhibit a sensitivity to co-bubbles. We identify 84 co-bubble connections, with 62 being positive and 22 negative. On average, each sector experiences the influence of more than 6 other sectors when price co-bubbles are considered. Notably, only the advanced materials and healthy living sectors exhibit minimal dependence on price bubbles in other sectors.

Overall, this study highlights evidence of price bubbles and co-bubbles in the new green economy market. Essentially, the presence of price bubbles in one sector (region), significantly affects the likelihood of price bubbles in another sector (region). The main green economy market index shows resilience to price bubbles in other sub-sectors. Interestingly, there is a resemblance in co-bubbles between the main green economy market index and the wind index. Price bubble in Asia shapes the price bubbles in the green economy market index of Europe and the USA.

Considering these insights, investors can use evidence of co-bubbles in the green economy market when navigating between sectors (Ang et al., 2005). Our findings are valuable when considering that the number of price bubble periods in the green economy market is much smaller than in other markets, for example, the food commodity market (Etienne et al., 2014), crude oil futures (Chang, 2024), and the coal market (Wang et al., 2024). We detect price bubbles in the main green economy market index and correlate it with the COVID-19 outbreak and subsequent pandemic development. Confirmation of bubble phases for the main green economy market index around the COVID-19 outbreak is also given by Basse et al. (2023). Furthermore, the clear connections between the indexes analysed, geographical regions, and sectors, are explained by the increasing role of investing in sustainable development companies (Mushafiq et al., 2023). Institutional and individual investors are generally interested in green economy market companies and tend to keep some part of their investment portfolio in shares of these companies. Maintaining constant shares in such companies in an investment portfolio can induce interactions leading to the occurrence of price bubbles. This is possible, as research indicates a positive correlation between environmental friendly solutions and the financial returns of large firms (Feng and Yuan, 2024).

5. Robustness analysis

To check the robustness of our results, we analyse the total return version of the green economy market indexes. If price bubbles occur in indexes it also should occur in total return indexes which contain dividends. Any additional cash flows should enhance the price bubble and co-bubble behaviours which represent a main part of our analysis. The results of the robustness analysis are reported in Appendix C (Table C.1). They show no major differences between the presence of price bubbles in the main indexes and geographical regions. Some minor differences are noted between price and total return indexes for the clean energy, developer/operator and renewable energy sectors. In these sectors, the difference in duration of price bubbles and length of the analysed time series is over 4 percentage points. For these three sectors, longer and more numerous price bubble periods are found when analysing total return indexes. Notably, based on price indexes, the average duration of the designated price bubbles is shorter. In general, it can be concluded that the choice of index (price or total return) for the sectors considered has no impact on the main results obtained regarding price bubble periods. However, the small differences, e.g. for the average duration of diagnosed price bubbles, are always higher for total return indexes.

The results of the co-bubble analysis based on logistic regression for the total return index are presented in Appendix D (Tables D.1, D.2, and D.3). The results for the main categories of indexes are similar in the value of the estimated coefficients and identical in the direction of co-movements. The overall number of observed connections is also identical. For the geographical regions, the results also remain consistent. For the sectoral analysis of the total return indexes, we obtain 82 statistically significant results compared to the results involving price indexes. Overall, the results remain the same as our main findings, providing further support for the reliability of the findings. This is in line with the rationale that for total return indexes, it is easier for price bubbles and co-bubbles to manifest, and this is why for the main empirical analysis we use price indexes.

6. Conclusion

This study analyses price bubbles and co-bubbles in the green economy market. The results show significant price bubbles in the green economy market, particularly in the sub-sector of the green economy market including solar and wind. However, none of the price bubbles detected receives an "ongoing" status at the end of the analysis. It should be emphasized that the most significant and longest price bubbles appears in the market due to the COVID-19 pandemic and Russian aggression against Ukraine in 2022. The same regularity is shown in previous studies for oil markets (Ajmi et al., 2021), the natural resources market in the context of increased short-term market volatility, the natural gas market (Akcora and Kocaaslan, 2023), the diamond market (Potrykus, 2022), and the cryptocurrency market (Bouri et al., 2019), as well as selected traditional or alternative investment markets (Potrykus, 2023). Therefore, it cannot be said that the green economy market, understood as a whole, is different from the investments mentioned above in terms of the causes of periods of price bubbles. Importantly, however, it is possible to identify narrow sectors of this market that are resistant to these phenomena (GFC, COVID-19, Russia's war against Ukraine) which, despite their occurrence, do not cause price bubbles for these sectors. Of the green economy market categories examined, the solar category is most susceptible to price bubble, and the water category is identified as the least sensitive. The total duration of price bubbles for the green economy market indexes also turns out to be much shorter than similar studies of this type conducted in other markets. So, even if the "painted green" phenomenon is taken into account, for right now, it cannot be said that this induces a price bubble pattern on the green economy market.

There is a bigger tendency for periods of price bubbles in the USA market than twin companies from the clean energy focus and green economy market families in Asia or Europe. The only region without statistically significant price bubbles is the USA clean energy region. Previous research into capital markets also shows a bigger tendency for price bubbles in the USA or Japanese markets than selected European countries. Moreover, there are sectors where no price bubbles are found: green IT index, bio/clean fuels, energy efficiency, smart grid, lighting, and energy management. Adding companies from these sectors to an investment portfolio should contribute to stabilizing its parameters, such as the rate of return and risk. This function should be particularly visible in times of turmoil in financial markets. This is an essential conclusion for people managing investment portfolios, investors, and market analysts. The second group distinguished in this research are sectors in which short-term and rare periods defined as price bubbles occur. This group includes companies from the advanced materials, green building, geothermal, healthy living, pollution mitigation, and energy storage sectors. The third group of companies, distinguished in the sector analysis, includes companies in which price bubbles occur most often and are characterized by a long duration compared to the entire sector, comprising the clean energy focused, developer/operator, fuel cell, natural resources, recycling, renewable energy generation, and green transportation sectors.

The findings reveal many statistically significant connections, which increase or decrease the probability of forming price bubble periods between the analysed indexes. For example, price bubbles in the solar index and global water index increase the probability of the appearance of price bubbles in the wind index. Price bubbles in the solar index reduce the presence of price bubbles in the global water index. The reverse connection is also true. The presence of price bubbles in Asia increase the probability of price bubble in Europe and the USA. We discover 84 connections in the sectoral analysis, 62 positive and 22 negative. However, the selection of the total return index or price index for analysis does not affect the outcomes regarding price bubbles and co-bubbles.

The above findings highlight important policy implications for investors, policymakers, and market analysts. Investors and portfolio managers should concentrate on sectors, such as green IT index, bio/clean fuels and energy efficiency that are less susceptible to price fluctuations, in order to control risk and stabilize their portfolios amid market turmoil. However, solar and wind are more vulnerable to price bubbles, and thus they need careful investment approaches, especially during times of economic or geopolitical unrest. Policymakers should focus on promoting innovation in resilient green technologies rather than implementing broad market interventions. Recognizing geographical differences, such as the higher price bubbles in USA compared to Europe and Asia, can further guide strategic investment decisions. While short-term investors may consider exploring sectors with temporary but frequent price bubbles, such as Geothermal and Advanced Materials, these sectors might present speculative opportunities but their inherent volatility calls for some caution. Further studies are needed in this regard.

CRediT authorship contribution statement

Marcin Potrykus: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Imran Ramzan:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Mazhar:** Writing – original draft, Investigation, Formal analysis. **Elie Bouri:** Writing – review & editing, Writing – original draft, Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix

A. Index description

Table A.1
Overview of index details

No.	Research cross-section	Price index name (abbreviation) used by NASDAQ	Description from https://indexes.nasdaqomx.com
1	Overall green finance index	QGREEN	The NASDAQ OMX Green Economy Index is a market-capitalization weighted index designed to track the performance of companies across the spectrum of industries most closely associated with the economic model around sustainable development through every economic sector. The index began on September 22, 2010 at a base value of 1000.00.
2	Main category indexes	GRNSOLAR	The NASDAQ OMX Solar Index is a sub sector index of the Green Economy Index designed to track companies that produce energy through solar power. The Solar Index rolls up to the primary sector Renewable Energy Generation Index.
3		GRNWATERL	The NASDAQ OMX Global Water Index is designed to track the performance of companies worldwide that are creating products that conserve and purify water for homes, businesses and industries. The index is weighted to enhance the underlying liquidity and increase the tradability of the index components.
4		GRNWIND	The NASDAQ OMX Wind Index is a sub sector index of the Green Economy Index designed to track companies that produce energy through wind power. The Wind Index rolls up to the primary sector Renewable Energy Generation Index.
5	Geographical indexes	GRNASIA	The NASDAQ OMX Green Economy Asia Index is a market-capitalization weighted designed to track the performance of companies across the spectrum of industries most closely associated with the economic model around sustainable development through every economic sector. The Green Economy Index includes companies involved with industries such as Energy Efficiency, Renewable Energy Generation, Advanced Materials, Green Building, and Healthy Living. The index is designed to serve as a global benchmark of all companies involved in the reduction of fossil-sourced fuels, products, services, and lifestyles of companies domiciled in the Asia.
6		GRNEUROPE	The same description as for GRNASIA but domiciled in the Europe.
7		GRNUS	The same description as for GRNASIA but domiciled in the United States of America.
8		GRNEXUS	The same description as for GRNASIA but domiciled outside the USA.
9		GRNFOCAS	The NASDAQ OMX Clean Energy Focused Asia Index is designed to track the sectors of the Green Economy specifically enabling the advancement of energy generation via non-fossil based sources. The index is comprised of the following sectors: Renewable Energy, Energy Efficiency, Advanced Materials, and Bio/Clean Fuels. Companies must be domiciled in Asia for inclusion into the index.
10		GRNFOCEUR	The same description as for GRNFOCAS but companies must be domiciled in Europe for inclusion into the index.
11		GRNFOCUS	The same description as for GRNFOCAS but companies must be domiciled in the United States of America for inclusion into the index.

(continued on next page)

Table A.1 (*continued*)

No.	Research cross-section	Price index name (abbreviation) used by NASDAQ	Description from https://indexes.nasdaqomx.com
12		GRNFCEXUS	The same description as for GRNFOCAS but companies must be domiciled outside the United States of America for inclusion into the index.
13	Sector indexes	GREENIT	The NASDAQ OMX Green IT is a sub sector index of the Green Economy Index designed to track companies that provide solutions that help companies decrease energy consumption by enabling collaboration online, efficient data centres, computer networks, and virtualization software. The Green IT Index rolls up to the broader Energy Efficiency Index in the Green family.
14		GRNAM	The NASDAQ OMX Advanced Materials Index is a primary sector index of the Green Economy Index designed to track companies producing materials with advanced properties that enable renewable technologies or reduce dependencies for petroleum-based products.
15		GRNBIO	The NASDAQ OMX Bio/Clean Fuels Index is a primary sector index of the Green Economy Index designed to track companies that manufacture fuels from plant-based material that replaces petroleum-based fuels for transportation.
16		GRNCLNFO	The NASDAQ OMX Clean Energy Focused Index is designed to track the sectors of the Green Economy specifically enabling the advancement of energy generation via non-fossil based sources. The index is comprised of the following sectors: Renewable Energy, Energy Efficiency, Advanced Materials, and Bio/Clean Fuels.
17		GRNDO	The NASDAQ OMX Developer/Operator Index is a sub sector index of the Green Economy Index designed to track companies that build and operate renewable energy projects such as solar and wind farms. The Developer/Operator Index rolls up to the primary sector Renewable Energy Generation Index.
18		GRNENEF	The NASDAQ OMX Energy Efficiency Index is a primary sector index of the Green Economy Index designed to track companies that dramatically increase energy efficiency inclusive of the subsectors Energy Management, Energy Storage, Smart Grid and Green IT.
19		GRNFUEL	The NASDAQ OMX Fuel Cell Index is a sub sector index of the Green Economy Index designed to track companies that produce energy through fuel cells. The Fuel Cell Index rolls up to the primary sector Renewable Energy Generation Index.
20		GRNGB	The NASDAQ OMX Green Building Index is a primary sector index of the Green Economy Index designed to track companies participating in advanced designs for retrofits and new buildings that lead to dramatic efficiency gains in energy and water consumption.
21		GRNGEO	The NASDAQ OMX Geothermal Index is a sub sector index of the Green Economy Index designed to track companies that produce energy through geothermal power. The Geothermal Index rolls up to the primary sector Renewable Energy Generation Index.
22		GRNGRD	The NASDAQ OMX Smart Grid Index is a sub sector index of the Green Economy Index designed to track companies that provide solutions resulting in modernizing the electric grid, making the grid more reliable, and more intelligent. The Smart Grid Index rolls up to the broader Energy Efficiency Index in the Green family.
23		GRNHL	The NASDAQ OMX Healthy Living Index is a primary sector index of the Green Economy Index designed to track companies that produce food, personal care and other products using benign, natural and certified organic ingredients based on sustainable agriculture practices.
24		GRNLIGHT	The NASDAQ OMX Lighting Index is a primary sector index of the Green Economy Index designed to track companies producing advanced lighting technologies resulting in dramatic efficiency gains in lighting through products such as compact fluorescents and LEDs.
25		GRNMAN	The NASDAQ OMX Energy Management is a sub sector index of the Green Economy Index designed to track companies that provide solutions that help companies reduce energy consumption, such as efficient motors, micro turbines, process controls, and appliances. The Energy Management Index rolls up to the broader Energy Efficiency Index in the Green family.
26		GRNNR	The NASDAQ OMX Natural Resources Index is a primary sector index of the Green Economy Index designed to track companies participating in the agriculture, forestry, and other natural products industries using sustainable methods such as certified organic agriculture and Forest Stewardship Certified forestry.
27		GRNPOL	The NASDAQ OMX Pollution Mitigation Index is a primary sector index of the Green Economy Index designed to track companies producing goods and services that reduce pollution from conventional industrial processes, power plants and other emitters.
28		GRNREC	The NASDAQ OMX Recycling Index is a primary sector index of the Green Economy Index designed to track companies participating in processes and products that greatly reduce the waste stream and extend useful life of materials from precious metals to asphalt by recovering, reusing and recycling them into next generation products.
29		GRNREG	The NASDAQ OMX Renewable Energy Generation index is a primary sector index of the Green Economy Index designed to track companies that produce energy through renewable sources such as solar, wind, geothermal, wave, and fuel cells.
30		GRNSTOR	The NASDAQ OMX Energy Storage is a sub sector index of the Green Economy Index designed to track companies that provide solutions increasing the ability of energy storage when needed, such as batteries. The Energy Storage Index rolls up to the broader Energy Efficiency Index in the Green family.
31		GRNTRN	The NASDAQ OMX Transportation Index is a primary sector index of the Green Economy Index designed to track companies focused on efficiency gains and pollution reduction associated with automobiles, trains, and other modes of transportation.

B. Descriptive statistics

Table B.1
Summary statistics of price indices

No.	Index	Mean	Median	Std. Dev.	Kur.	Ske.	Min.	Max.	JB Stat.
1	Main Green Economy Index	1591.7	1348.1	723.9	−0.1	1.0	533.7	3640.9	853.9 ***
2	Solar Index	1637.7	1156.6	1166.8	−0.1	1.1	284.7	4812.6	973.9 ***
3	Global Water Index	1269.3	1172.2	408.7	−0.2	0.8	493.5	2332.6	499.3 ***
4	Wind Index	1771.5	1604.5	881.4	0.2	0.8	365.5	5174.0	461.1 ***
5	Green Economy Asia	1042.9	1014.1	220.5	0.2	0.6	607.9	1794.1	278.3 ***
6	Green Economy Europe	1249.5	1202.5	314.3	−0.2	0.5	581.7	2126.3	223.8 ***
7	Green Economy USA	1808.8	1321.8	1182.1	0.0	1.2	453.2	5134.5	1098.6 ***
8	Green Economy Outside USA	1208.9	1183.8	282.1	0.1	0.5	559.5	2006.1	217.3 ***
9	Clean Energy Asia	819.8	738.4	287.4	1.6	1.1	367.2	2158.6	1511.5 ***
10	Clean Energy Europe	1222.0	1131.0	381.5	−0.6	0.7	584.0	2234.1	425 ***
11	Clean Energy USA	1616.0	1199.3	910.2	−0.7	0.9	471.0	3731.9	683.4 ***
12	Clean Energy Outside USA	1163.7	1065.6	336.1	−0.1	0.8	538.9	2154.5	475.7 ***
13	Advanced Materials	1277.2	1294.0	403.9	−0.9	0.1	444.2	2271.2	183.2 ***
14	Bio/Clean Fuels	1083.7	974.8	404.8	3.6	1.6	327.7	3368.0	4508.7 ***
15	Clean Energy	1394.8	1165.0	592.1	−0.5	0.9	510.5	2837.7	631.7 ***
16	Developer/Operator	1586.5	1350.5	576.4	−0.6	0.7	760.0	3211.0	493 ***
17	Energy Efficiency	1468.2	1244.1	708.7	−0.5	0.8	430.6	3406.6	503.7 ***
18	Energy Management	1389.0	1203.9	662.7	0.9	1.2	414.0	3783.9	1365.1 ***
19	Energy Storage	925.8	860.3	326.8	−0.2	0.7	412.6	1824.9	390.4 ***
20	Fuel Cell	2230.9	1164.8	2209.6	1.0	1.4	299.4	11,457.0	1697.6 ***
21	Geothermal	1043.3	989.9	236.3	0.9	0.9	510.0	2144.3	788.3 ***
22	Green Building	1328.3	1409.4	286.8	0.1	−0.8	378.2	1867.1	465.5 ***
23	Green IT	1521.7	1178.4	722.4	−1.1	0.6	499.3	3177.6	570.7 ***
24	Healthy Living	1609.0	1754.3	683.7	−0.8	0.0	364.3	3283.4	120.6 ***
25	Lighting	1132.2	1084.2	313.6	0.2	0.5	415.6	2054.7	225.9 ***
26	Natural Resources	1518.5	1193.5	911.9	0.3	1.3	375.2	3909.6	1327 ***
27	Pollution Mitigation	1436.0	1332.4	533.0	−0.5	0.6	529.5	2814.2	357.1 ***
28	Recycling	1788.0	1310.9	1033.4	0.0	1.1	462.6	4725.5	991.4 ***
29	Renewable Energy	1527.9	1246.1	682.7	−0.4	0.8	531.8	3514.6	519.7 ***
30	Smart Grid	1220.6	1187.4	497.8	0.5	0.7	313.0	3004.2	475.2 ***
31	Transportation	2217.6	1595.3	1863.6	1.0	1.5	513.1	9074.8	2009.4 ***

Notes: The table gives the descriptive statistics of the indexes under study. Kur. is for kurtosis, Ske. is for skewness and JB is the Jarque–Bera statistic.

C. GSADF test results

Table C.1
GSADF test procedure for the price and total return indexes

		Price index		Total return index	
No.	Index name	GSADF test statistic value	Symbol for significant level	GSADF test statistic value	Symbol for significant level
1	Main Green Economy Index	3.806	***	4.187	***
2	Solar Index	5.866	***	6.170	***
3	Global Water Index	3.321	***	3.287	***
4	Wind Index	6.343	***	6.852	***
5	Green Economy Asia	4.391	***	4.252	***
6	Green Economy Europe	2.827	**	2.821	**
7	Green Economy USA	4.703	***	5.151	***
8	Green Economy Outside USA	3.250	***	3.203	***
9	Clean Energy Asia	5.701	***	6.052	***
10	Clean Energy Europe	3.532	***	3.943	***
11	Clean Energy USA	1.898		2.434	*
12	Clean Energy Outside USA	3.564	***	4.037	***
13	Advanced Materials	3.343	***	3.382	***
14	Bio/Clean Fuels	1.930		1.930	
15	Clean Energy	2.708	**	3.062	***
16	Developer/Operator	4.756	***	4.704	***
17	Energy Efficiency	1.983		2.069	
18	Energy Management	1.825		2.280	
19	Energy Storage	8.325	***	8.456	***
20	Fuel Cell	17.105	***	17.110	***
21	Geothermal	3.241	***	3.261	***
22	Green Building	6.622	***	5.956	***
23	Green IT	1.954		2.292	
24	Healthy Living	3.426	***	3.415	***
25	Lighting	2.288		2.323	
26	Natural Resources	3.714	***	4.120	***
27	Pollution Mitigation	3.162	***	3.106	***

(continued on next page)

Table C.1 (*continued*)

		Price index		Total return index	
No.	Index name	GSADF test statistic value	Symbol for significant level	GSADF test statistic value	Symbol for significant level
28	Recycling	3.068	***	3.603	***
29	Renewable Energy	5.786	***	6.516	***
30	Smart Grid	1.728		2.464	*
31	Transportation	10.378	***	10.511	***

D. Logistic regression results for total return indexes

Table D.1
Logistic regression results of the green economy and its components

	Solar Index	Global Water Index	Wind Index
Solar Index		2.462 (0.187) ***	2.425 (0.140) ***
Global Water Index	2.462 (0.187) ***		0.686 (0.208) ***
Wind Index	2.425 (0.14) ***	0.686 (0.208) ***	
Const (β_0)	−3.315 (0.082) ***	−3.866 (0.106) ***	−2.888 (0.067) ***
McFadden R^2	0.207	0.157	0.132

Table D.2
Logistic regression results for the geographical indexes cross-section

	Green Economy Asia	Green Economy Europe	Green Economy USA	Clean Energy Asia	Clean Energy Europe
Panel A: Green Economy					
Green Economy Asia		4.756 (0.43) ***	1.612 (0.211) ***		
Green Economy USA	1.612 (0.211) ***				
Panel B: Clean Energy					
Clean Energy Asia					3.121 (0.200) ***
Clean Energy Europe				3.121 (0.200) ***	
Const (β0)	−3.917 (0.110) ***	−6.356 (0.354) ***	−2.528 (0.056) ***	−3.908 (0.106) ***	−3.689 (0.096) ***
McFadden R^2	0.025	0.343	0.018	0.143	0.127

Table D.3
Logistic regression results for sectoral indexes

	Advanced Materials	Clean Energy	Developer/ Operator	Energy Storage	Fuel Cell	Geothermal	Green Building	Healthy Living	Natural Resources	Pollution Mitigation	Recycling	Renewable Energy	Transportation
Advanced Materials							2.89 (0.615) ***	1.986 (0.641) ***				2.664 (0.582) ***	1.311 (0.67) *
Clean Energy			−1.222 (0.28) ***	2.399 (0.674) ***	1.623 (0.565) ***	3.19 (0.699) ***			1.191 (0.33) ***		5.623 (0.557) ***	5.173 (0.499) ***	1.862 (0.351) ***
Developer/ Operator					1.233 (0.313) ***		2.477 (0.306) ***		1.318 (0.268) ***		0.667 (0.269) **	3.023 (0.19) ***	1.952 (0.232) ***
Energy Storage	1.353 (0.795) *				2.919 (0.429) ***	3.663 (0.722) ***			2.79 (0.507) ***				
Fuel Cell	−1.786 (0.664) ***	2.197 (0.461) ***	2.003 (0.263) ***	2.199 (0.423) ***					0.86 (0.465) *		−8.928 (0.723) ***	3.215 (0.408) ***	
Geothermal				1.902 (0.718) ***				2.609 (0.679) ***		7.732 (0.698) ***	3.196 (0.661) ***	2.398 (1.17) **	−1.769 (0.955) *

(*continued on next page*)

Table D.3 (*continued*)

	Advanced Materials	Clean Energy	Developer/ Operator	Energy Storage	Fuel Cell	Geothermal	Green Building	Healthy Living	Natural Resources	Pollution Mitigation	Recycling	Renewable Energy	Transportation
Green Building	1.472 (0.685) **		2.481 (0.285) ***					2.235 (0.359) ***			2.259 (0.336) ***	1.476 (0.377) ***	−3.159 (0.72) ***
Healthy Living	1.755 (0.741) **		−3.244 (0.774) ***		2.081 (0.66) ***		2.764 (0.372) ***		1.737 (0.655) ***				
Natural Resources	−1.042 (0.522) **		0.595 (0.249) **	2.428 (0.471) ***	−1.043 (0.593) *			1.062 (0.539) **			4.316 (0.384) ***	2.715 (0.52) ***	1.007 (0.338) ***
Pollution Mitigation						8.145 (0.968) ***			−3.177 (0.719) ***				−5.462 (1.199) ***
Recycling		6.08 (0.523) ***	0.499 (0.229) **	−1.587 (0.46) ***	−4.313 (0.582) ***		1.355 (0.323) ***		4.396 (0.367) ***			−3.504 (0.506) ***	4.167 (0.279) ***
Renewable Energy	2.327 (0.545) ***	4.569 (0.405) ***	2.827 (0.176) ***	−2.913 (0.709) ***	3.237 (0.415) ***		0.786 (0.346) **	−1.831 (0.593) ***	1.607 (0.337) ***		−3.959 (0.504) ***		4.159 (0.277) ***
Transportation	1.365 (0.547) **	1.676 (0.33) ***	0.892 (0.221) ***		2.16 (0.3) ***		−3.186 (0.523) ***				4.625 (0.309) ***	3.585 (0.316) ***	
Const (β0)	−6.01 (0.297) ***	−8.973 (0.584) ***	−3.091 (0.077) ***	−3.731 (0.104) ***	−5.943 (0.276) ***	−8.115 (0.701) ***	−5.048 (0.191) ***	−3.999 (0.115) ***	−6.615 (0.333) ***	−7.357 (0.578) ***	−3.734 (0.105) ***	−4.231 (0.13) ***	−5.455 (0.223) ***
McFadden R2	0.195	0.788	0.375	0.116	0.620	0.697	0.219	0.065	0.655	0.650	0.579	0.679	0.722

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